

## Query evaluation – Combined, data, query complexity

### Theorem (Combined complexity of query evaluation)

The complexity of  $\{\langle \mathcal{I}, \alpha, \varphi \rangle \mid \mathcal{I}, \alpha \models \varphi\}$  is:

- *time*: exponential
- *space*: PSPACE-complete — see [Var82] for hardness

### Theorem (Data complexity of query evaluation)

The complexity of  $\{\langle \mathcal{I}, \alpha \rangle \mid \mathcal{I}, \alpha \models \varphi\}$  is:

- *time*: polynomial
- *space*: LOGSPACE

### Theorem (Query complexity of query evaluation)

The complexity of  $\{\langle \alpha, \varphi \rangle \mid \mathcal{I}, \alpha \models \varphi\}$  is:

- *time*: exponential
- *space*: PSPACE-complete — see [Var82] for hardness

## Example

We consider FOL interpretations exactly as used in relational databases. This requires to drop functions except for constants. Moreover we assume that the interpretation of constants is the identity function, that is constants are interpreted as themselves. This allows us to drop also the interpretation of constants from our interpretations, which now have the form:

$$\mathcal{I} = (\Delta^{\mathcal{I}}, P_1^{\mathcal{I}}, P_2^{\mathcal{I}}, \dots, P_n^{\mathcal{I}}).$$

**Interpretation:**  $\mathcal{I}$  is as follows (also given in relational notation):

- $\Delta^{\mathcal{I}} = \{\text{john}, \text{paul}, \text{george}, \text{mick}, \text{ny}, \text{london}, 0, 1, \dots, 100\}$
- $\text{Person}^{\mathcal{I}} = \{(\text{john}, 30), (\text{paul}, 60), (\text{george}, 35), (\text{mick}, 35)\}$
- $\text{Lives}^{\mathcal{I}} = \{(\text{john}, \text{ny}), (\text{paul}, \text{ny}), (\text{george}, \text{london}), (\text{mick}, \text{london})\}$
- $\text{Manages}^{\mathcal{I}} = \{(\text{paul}, \text{john}), (\text{george}, \text{mick}), (\text{paul}, \text{mick})\}$

$\text{Person}^{\mathcal{I}}$

| name   | age |
|--------|-----|
| john   | 30  |
| paul   | 60  |
| george | 35  |
| mick   | 35  |

$\text{Lives}^{\mathcal{I}}$

| name   | city   |
|--------|--------|
| john   | ny     |
| paul   | ny     |
| george | london |
| mick   | london |

$\text{Manages}^{\mathcal{I}}$

| boss   | emp. name |
|--------|-----------|
| paul   | john      |
| george | mick      |
| paul   | mick      |

**Query:** find name and age of persons who live in the same city as their boss.

$$\exists z, w. \text{Person}(x, y) \wedge \text{Manages}(z, x) \wedge \text{Lives}(x, w) \wedge \text{Lives}(z, w)$$

## Example - Interpretation

Consider the following interpretation  $\mathcal{I}$ :

- $\Delta^{\mathcal{I}}$  is equal to the *active domain*: all objects occurring in any predicate extension.
- $Sailors^{\mathcal{I}}$  see table below
- $Boats^{\mathcal{I}}$  see table below
- $Reserves^{\mathcal{I}}$  see table below

$Sailors^{\mathcal{I}}$

| sid | sname  |
|-----|--------|
| 22  | dustin |
| 31  | lubber |
| 58  | rusty  |

$Boats^{\mathcal{I}}$

| bid | color |
|-----|-------|
| 101 | red   |
| 102 | green |
| 103 | red   |
| 104 | blue  |

$Reserves^{\mathcal{I}}$

| sid | bid | day      |
|-----|-----|----------|
| 22  | 101 | 10/10/96 |
| 58  | 103 | 11/12/96 |

## Example - Queries

- Find the names of the sailors who have reserved boat 103.
- Find the names of the sailors who have reserved a red boat.
- Find the colors of the boats reserved by Bob.
- Find the names of the sailors who have reserved at least one boat.
- Find the names of the sailors who have reserved a red and a green boat.
- Find the names of the sailors who have reserved a red or a green boat.
- Find the names of the sailors who have reserved at least two boats.
- Find the names of the sailors who have not reserved a red boat.
- Find the names of the sailors who have reserved all boats.
- Find the names of the sailors who have reserved all red boats.

## Example - Queries

- Find the names of the sailors who have reserved boat 103.

$$\exists x.Sailors(x, y) \wedge \exists z.Reserves(x, 103, z)$$

$$q(y) \leftarrow Sailors(x, y), Reserves(x, 103, z)$$

- Find the names of the sailors who have reserved a red boat.

$$\exists x.Sailors(x, y) \wedge \exists z, w.Reserves(x, z, w) \wedge Boats(z, red)$$

$$q(y) \leftarrow Sailors(x, y), Reserves(x, z, w), Boats(z, red)$$

- Find the colors of the boats reserved by Bob.

$$\exists x.Boats(x, y) \wedge \exists z, w.Reserves(z, y, w) \wedge Sailor(z, Bob)$$

$$q(y) \leftarrow Boats(x, y), Reserves(z, y, w), Sailor(z, Bob)$$

- Find the names of the sailors who have reserved at least one boat.

$$\exists x.Sailors(x, y) \wedge \exists z, w.Reserves(x, z, w)$$

$$q(y) \leftarrow Sailors(x, y), Reserves(x, z, w)$$

- Find the names of the sailors who have reserved a red and a green boat.

$$\exists x.S(x, y) \wedge \exists z, w.R(x, z, w) \wedge B(z, red) \wedge \exists z', w'.R(x, z', w') \wedge B(z', green)$$

$$q(y) \leftarrow Sailors(x, y), Reserves(x, z, w), Boats(s, red), Reserves(x, z', w'), Boats(z', green)$$

## Example - Queries

- Find the names of the sailors who have reserved a red or a green boat.

$$\exists x. \text{Sailors}(x, y) \wedge \exists z, w. \text{Reserves}(x, z, w) \wedge (\text{Boats}(z, \text{red}) \vee \text{Boats}(z, \text{green}))$$

$$\begin{aligned} q(y) &\leftarrow \text{Sailors}(x, y), \text{Reserves}(x, z, w), \text{Boats}(z, \text{red}) \\ q(y) &\leftarrow \text{Sailors}(x, y), \text{Reserves}(x, z, w), \text{Boats}(z, \text{green}) \end{aligned}$$

- Find the names of the sailors who have reserved at least two boats.

$$\exists x. \text{Sailors}(x, y) \wedge \exists z, w. \text{Reserves}(x, z, w) \wedge \exists z', w'. \text{Reserves}(x, z', w') \wedge z \neq z'$$

- Find the names of the sailors who have not reserved a red boat.

$$\exists x. \text{Sailors}(x, y) \wedge \forall z, w. (\text{Reserves}(x, z, w) \rightarrow \neg \text{Boats}(z, \text{red}))$$

- Find the names of the sailors who have reserved all boats.

$$\exists x. \text{Sailors}(x, y) \wedge \forall z, c. (\text{Boats}(z, c) \rightarrow \exists w. \text{Reserves}(x, z, w))$$

- Find the names of the sailors who have reserved all red boats.

$$\exists x. \text{Sailors}(x, y) \wedge \forall z. (\text{Boats}(z, \text{red}) \rightarrow \exists w. \text{Reserves}(x, z, w))$$

# References

[Var82] M. Y. Vardi.

The complexity of relational query languages.

In *Proc. of the 14th ACM SIGACT Symp. on Theory of Computing (STOC'82)*, pages 137–146, 1982.

# Formal proof systems for propositional logic

## Hilbert proof system

Chiara Ghidini – Maurizio Lenzerini

FBK-IRST, Trento, Italy – Sapienza Università di Roma

Revised and adapted by F. Patrizi



## Formal proofs are at the core of modern logic

- The modern notion of **symbolic formal proof** was developed in the 20th century by logicians and mathematicians such as Russell, Frege and Hilbert.
- The benefit of symbolic formal proofs is that they are based on **pure syntax, i.e., symbolic manipulation**: *a precisely defined symbolic formalism with procedures for transforming statements into other statements, based solely on their **form**.*
- **No intuition or interpretation is needed**, merely applications of agreed upon rules to a set of agreed upon formulas.

# The notion of formal system

Intuitively, a **formal system** consists of a language over some alphabet of symbols together with initial strings and production rules that are used to generate some of the strings in the language. More precisely,

## Formal system

A formal system has the following components:

- An alphabet of symbols.
- A **syntax** that defines which sequences of symbols are in the language of our formal system.
- A decidable set of **initial sequences**.
- A finite set of **production rules** from which the set of **conclusions** of the system is generated (each conclusion is generated through a finite set of application of the production rules on both the initial sequences and the already generated conclusions).

# Proof system

## Formal proof system

A symbolic formal proof system (or simply **proof system**)  $\Sigma$  for propositional logic is constituted by:

- The set of **axioms** of  $\Sigma$ , i.e., a set of propositional formulas such that there is an effective procedure to decide if a given formula is an axiom.
- The (finite) set  $R_1, \dots, R_n$  of **inference rules** of  $\Sigma$ , where each inference rule  $R_i$  is a relation that associates to a set of  $j_i$  formulas  $A_1, \dots, A_{j_i}$ , one formula  $A$  that is called the **direct consequence of  $A_1, \dots, A_{j_i}$  through  $R_i$** .

A formula  $A$  that is a direct consequence of  $A_1, \dots, A_j$  through  $R$  is said to be **derived from  $A_1, \dots, A_j$  by  $R$** . When we want to describe an inference rule  $R$  applied to  $A_1, \dots, A_j$  to derive  $A$ , we write:

$$R \quad \frac{A_1, \dots, A_j}{A}$$

# Proof, provability, theorem

## Proof

A **proof in  $\Sigma$**  is a finite sequence of formulas  $\phi_1, \dots, \phi_n$  such that each  $\phi_k$  is either

- an axiom of  $\Sigma$ , or
- a direct consequence of a set of previous formulas in the sequence through an inference rule of  $\Sigma$ .

## Proof of $\phi$

A **proof of  $\phi$  in  $\Sigma$**  is a proof  $\phi_1, \dots, \phi_n$  in  $\Sigma$  such that  $\phi_n = \phi$ .

## Theorem

A formula  $\phi$  is **provable in  $\Sigma$**  if there is a proof of  $\phi$  in  $\Sigma$ , and in this case we write  $\vdash_{\Sigma} \phi$  (or simply  $\vdash \phi$  when  $\Sigma$  is understood), and we also say that  $\phi$  is a **theorem** in (or of)  $\Sigma$ .

# Proof from hypotheses

## Proof from hypothesis

A **proof of  $\phi$  in  $\Sigma$  from a set of hypothesis (or assumptions)  $\Gamma$**  is a sequence of formulas

$\phi_1, \dots, \phi_n$ , with  $\phi_n = \phi$ , such that each  $\phi_k$  is either

- an axiom of  $\Sigma$ , or
- an **hypothesis** in  $\Gamma$ , or
- a direct consequence of a set of previous formulas in the sequence through an inference rule of  $\Sigma$ .

## Inference or derivation

We say that  **$\phi$  is inferred (or derived) from  $\Gamma$  in  $\Sigma$** , written  $\Gamma \vdash_{\Sigma} \phi$  (or simply  $\Gamma \vdash \phi$  when  $\Sigma$  is understood), if there is a proof of  $\phi$  in  $\Sigma$  from  $\Gamma$ .

## Consistency

We say that a set  $\Gamma$  of formulas is  **$\Sigma$ -consistent** if no formula  $\phi$  exists such that both  $\Gamma \vdash_{\Sigma} \phi$  and  $\Gamma \vdash_{\Sigma} \neg\phi$ , otherwise is  **$\Sigma$ -inconsistent**.

## Properties of a proof system

Analogously to logical implication, it is interesting to check whether the following properties hold for a certain  $\vdash_{\Sigma}$ :

- **Reflexivity** (If  $A \in \Gamma$ , then  $\Gamma \vdash_{\Sigma} A$ )
- **Ex falso sequitur quodlibet** (If  $\Gamma$  is unsatisfiable, then  $\Gamma \vdash_{\Sigma} A$  for all  $A$ )
- **Monotonicity** (If  $\Gamma \vdash_{\Sigma} A$  then  $\Gamma \cup \Delta \vdash_{\Sigma} A$ )
- **Cut** (If  $\Gamma \vdash_{\Sigma} A$  and  $\Delta \cup \{A\} \vdash_{\Sigma} B$  then  $\Gamma \cup \Delta \vdash_{\Sigma} B$ )
- **Compactness** (If  $\Gamma \vdash_{\Sigma} A$ , then there is a finite subset  $\Gamma_0 \subseteq \Gamma$ , such that  $\Gamma_0 \vdash_{\Sigma} A$ )
- **Deduction principle** ( $\Gamma \cup \{A\} \vdash_{\Sigma} B$  if and only if  $\Gamma \vdash_{\Sigma} A \rightarrow B$ )
- **Refutation principle** ( $\Gamma \vdash_{\Sigma} A$  if and only if  $\Gamma \cup \{\neg A\}$  is inconsistent)
- **Consistency**:  $\vdash_{\Sigma}$  is consistent if there is no formula  $A$  such that both  $\vdash_{\Sigma} A$  and  $\vdash_{\Sigma} \neg A$

# Hilbert formal proof system for propositional logic

We study the **Hilbert proof system**, that we denote by  $\mathcal{H}$ . In what follows,  $\phi$  and  $\psi$  denote formulas.

## Axioms of $\mathcal{H}$

$$\mathbf{A1} \quad \phi \rightarrow (\psi \rightarrow \phi)$$

$$\mathbf{A2} \quad (\phi \rightarrow (\psi \rightarrow \theta)) \rightarrow ((\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \theta))$$

$$\mathbf{A3} \quad (\neg\psi \rightarrow \neg\phi) \rightarrow ((\neg\psi \rightarrow \phi) \rightarrow \psi)$$

## Inference rule(s) of $\mathcal{H}$

$$\mathbf{MP} \quad \frac{\phi, \phi \rightarrow \psi}{\psi}$$

## Why there are no axioms for $\wedge$ and $\vee$ and $\equiv$ ?

Formulas with connectives  $\wedge$  and  $\vee$  are rewritten into equivalent formulas containing only  $\rightarrow$  and  $\neg$ .

$$A \wedge B \quad \equiv \quad \neg(A \rightarrow \neg B)$$

$$A \vee B \quad \equiv \quad \neg A \rightarrow B$$

$$A \equiv B \quad \equiv \quad \neg((A \rightarrow B) \rightarrow \neg(B \rightarrow A))$$